

Management of Type 2 Diabetes Mellitus in Adults

A Clinical Practice Guideline for Primary Care Settings

Illustrative / synthetic document created for training and demonstration purposes (e.g., testing retrieval-augmented generation systems). Not a substitute for official clinical guidance, and not intended for use in real patient care.

1. Purpose and Scope

This guideline provides a structured, evidence-informed approach to the diagnosis, initial management, ongoing monitoring, and referral of adults (aged 18 years and older) with Type 2 Diabetes Mellitus (T2DM) in a primary care setting. It does not cover gestational diabetes, Type 1 diabetes, or diabetes management in pediatric populations.

The guideline is intended for use by general physicians, nurse practitioners, and allied health professionals involved in the routine management of adults with T2DM.

2. Diagnostic Criteria

A diagnosis of Type 2 Diabetes Mellitus should be confirmed using one of the following criteria, ideally with a repeat confirmatory test on a separate day unless unequivocal hyperglycemia is present:

Test	Threshold	Fasting State	Notes
Fasting Plasma Glucose (FPG)	≥ 126 mg/dL (7.0 mmol/L)	No caloric intake for ≥ 8 hours	Preferred first-line test
Oral Glucose Tolerance Test (2-hr, 75g load)	≥ 200 mg/dL (11.1 mmol/L)	Fasting before load	Useful when FPG is borderline
HbA1c	≥ 6.5%	Not required	Should use a standardized, validated assay
Random Plasma Glucose	≥ 200 mg/dL with classic symptoms	Not required	Symptoms: polyuria, polydipsia, unexplained weight loss

Prediabetes is defined as FPG 100–125 mg/dL, 2-hour OGTT 140–199 mg/dL, or HbA1c 5.7–6.4%. Patients meeting prediabetes criteria should be counselled on lifestyle modification and rescreened annually.

3. Initial Assessment

At the time of diagnosis, the following should be assessed and documented:

- Baseline HbA1c, fasting lipid profile, renal function (serum creatinine, eGFR), and urine albumin-to-creatinine ratio (ACR)
- Blood pressure, measured on at least two occasions
- Body mass index (BMI) and waist circumference
- Comprehensive foot examination, including monofilament testing for peripheral neuropathy
- Dilated eye examination or retinal photography, with referral to ophthalmology if not performed in the preceding 12 months
- Screening for depression and diabetes-related distress using a validated tool
- Review of vaccination status (influenza, pneumococcal, hepatitis B where indicated)

4. Glycemic Targets

Patient Profile	Target HbA1c	Target FPG	Rationale
Healthy adult, recently diagnosed, low hypoglycemia risk	< 6.5% – 7.0%	80–130 mg/dL	Maximize long-term complication reduction
Older adult with comorbidities or hypoglycemia risk	7.5% – 8.0%	90–150 mg/dL	Balance benefit against hypoglycemia and falls risk
Limited life expectancy / frailty	< 8.5%	100–180 mg/dL	Prioritize symptom control and safety

5. Pharmacologic Management

5.1 First-Line Therapy

Metformin, titrated to a maximally tolerated dose (typically up to 2000 mg/day in divided doses), remains first-line therapy for most adults with T2DM unless contraindicated (eGFR < 30 mL/min/1.73m², acute illness with risk of lactic acidosis, or documented intolerance).

5.2 Second-Line Agents

Selection of a second agent should be guided by the presence of established cardiovascular disease, chronic kidney disease, heart failure, weight considerations, and cost/access. Commonly used classes include:

- SGLT2 inhibitors — preferred when atherosclerotic cardiovascular disease, heart failure, or chronic kidney disease is present
- GLP-1 receptor agonists — preferred when weight reduction is a priority or established cardiovascular disease is present
- DPP-4 inhibitors — considered when a low risk of hypoglycemia and weight neutrality are desired
- Sulfonylureas — effective and low-cost, but carry higher hypoglycemia risk
- Basal insulin — initiated when HbA1c remains substantially above target despite combination oral therapy, or in symptomatic hyperglycemia

5.3 Escalation Principle

Therapy should be reassessed every 3 months until glycemic targets are met, then at least every 6 months thereafter. Failure to reach target HbA1c within 3 months of a therapy change should prompt treatment intensification rather than prolonged observation.

6. Monitoring Schedule

Parameter	Frequency	Frequency (if uncontrolled)	Notes
HbA1c	Every 6 months	Every 3 months	Use point-of-care testing where available
Renal function / ACR	Annually	Every 6 months	More frequent if on SGLT2i or ACEi/ARB titration
Lipid profile	Annually	As clinically indicated	Consider statin therapy per cardiovascular risk
Foot examination	Annually	Every visit if high-risk foot	Includes monofilament and pulse assessment

Eye examination	Annually	Per ophthalmology advice	Can extend to 2 years if prior exams normal
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7. Referral Criteria

Refer to endocrinology or specialist diabetes services when any of the following apply:

- HbA1c remains > 9% despite triple therapy for 6 months
- Suspected Type 1 diabetes or Latent Autoimmune Diabetes in Adults (LADA)
- Recurrent severe hypoglycemia or hypoglycemia unawareness
- eGFR < 30 mL/min/1.73m² or rapidly declining renal function
- Diabetic foot ulceration or suspected Charcot foot
- Planning for pregnancy or newly diagnosed diabetes in pregnancy
- Need for insulin pump therapy or continuous glucose monitoring assessment

8. Lifestyle and Self-Management Education

All patients should be offered structured diabetes self-management education at diagnosis and at key transition points (e.g., insulin initiation). Core topics include:

- Carbohydrate-aware meal planning and portion guidance
- At least 150 minutes per week of moderate-intensity physical activity, individualized to functional capacity
- Blood glucose self-monitoring technique and interpretation, where applicable
- Sick-day rules and hypoglycemia recognition/management
- Smoking cessation support where relevant

9. Documentation

Each visit should document current glycemic control, medication adherence and tolerability, screening status for microvascular and macrovascular complications, and any changes to the management plan. A structured diabetes flow sheet or registry entry is recommended to support continuity of care across providers.

10. Cardiovascular Risk Reduction

Type 2 diabetes is an independent risk factor for atherosclerotic cardiovascular disease, and comprehensive management should extend beyond glycemic control to address overall cardiovascular risk.

- Blood pressure target of < 130/80 mmHg for most adults with T2DM, individualized for frailty and hypotension risk
- Moderate- to high-intensity statin therapy for most adults aged 40–75 years with T2DM, regardless of baseline LDL, unless contraindicated
- Low-dose aspirin (75–100 mg/day) for secondary prevention in patients with established cardiovascular disease; not routinely recommended for primary prevention in most patients
- Smoking cessation counselling at every visit for current smokers
- Consider SGLT2 inhibitors or GLP-1 receptor agonists with proven cardiovascular benefit in patients with established atherosclerotic disease or heart failure, independent of glycemic control needs

11. Special Populations

11.1 Older Adults

Management should prioritize avoidance of hypoglycemia and polypharmacy over strict glycemic targets, particularly in patients with frailty, cognitive impairment, or limited life expectancy. Simplify regimens where possible and reassess the ongoing necessity of each medication at every visit.

11.2 Chronic Kidney Disease

Metformin dose should be reduced when eGFR falls between 30–45 mL/min/1.73m² and discontinued below 30 mL/min/1.73m². SGLT2 inhibitors have demonstrated renal-protective effects and may be continued at lower eGFR thresholds per current product labeling; consult specific prescribing information.

11.3 Pregnancy and Preconception

Women with T2DM who are planning pregnancy should be transitioned from oral agents (with the general exception of metformin, used selectively) to insulin therapy prior to conception where feasible, and referred for preconception counselling and folic acid supplementation.

12. Common Comorbidities and Screening

Comorbidity	Screening Tool	Frequency	Action if Positive
Depression	PHQ-9 or equivalent	Annually	Refer to behavioral health; consider impact on self-management
Peripheral neuropathy	10g monofilament, vibration sense	Annually	Foot care education, referral to podiatry if high-risk
Nonalcoholic fatty liver disease	Liver enzymes, consider FIB-4 score	At diagnosis, then periodically	Refer to hepatology if high fibrosis risk
Obstructive sleep apnea	Clinical screening questionnaire	At diagnosis and if symptomatic	Refer for sleep study if suggestive symptoms

13. Illustrative Case Scenario

The following synthetic vignette is included to illustrate application of this guideline and is not derived from a real patient.

A 54-year-old individual presents with fasting plasma glucose of 158 mg/dL on two separate occasions and HbA1c of 8.1%. BMI is 31 kg/m², blood pressure is 138/86 mmHg, and there is no known history of cardiovascular or renal disease. Initial management includes lifestyle counselling, initiation of metformin titrated to 1000 mg twice daily, and a statin given age and diabetes status. Follow-up HbA1c at 3 months is 7.2%; the plan is continued with reassessment at 6 months, alongside baseline retinal and foot examinations and annual laboratory screening as outlined in Sections 3 and 6.

Appendix A: Summary Treatment Algorithm

- Step 1: Lifestyle modification + Metformin (if not contraindicated)
- Step 2: Add second agent based on comorbidity profile (SGLT2i, GLP-1 RA, DPP-4i, or sulfonylurea)
- Step 3: Add third agent from a different class, or consider basal insulin
- Step 4: Intensify insulin regimen (basal-bolus) or refer to specialist care

End of document. Document ID: MED-CPG-001 (synthetic training corpus).